

Local Indistinguishability

Global Obstruction and Minimal Predictive Memory

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Abstract

A precise reporter-boundary theorem is developed: a global label fails to factor through a family of local reporters exactly when there exist locally indistinguishable but globally distinguishable configurations. This biconditional directly realizes the defining local/global structure of topological order. The result is separated from a broader analogy between self-reference obstruction and nontrivial bundles, because the latter currently lacks quantization and deformation stability. Classical future-equivalence classes give deterministic predictive-memory lower bounds, while nonorthogonal quantum memory can compress the same process. Marginals of a bipartite state omit a calculable number of joint-correlation coordinates, so compatible marginals do not determine the operative relation. A graded dictionary distinguishes exact, structural, and heuristic transfers and identifies the construction of a deformation-stable reflexive invariant as the principal open problem.

Keywords: local indistinguishability; topological order; predictive memory; factorization; bond dimension; obstruction

Scope and status: The factorization theorem, coordinate count, and classical predictive-memory statements are exact. Topological analogies, quantization, stability, and the proposed invariant remain separate open questions.

Suggested citation: Singh, Abhijit. 2026. *Local Indistinguishability, Global Obstruction, and Minimal Predictive Memory*. Research manuscript, version 1.0.

Document status: Research manuscript, version 1.0, August 2026.

1 Reporter boundary

Let Σ be a set of configurations, $J : \Sigma \rightarrow O$ a global label, and $K_1, \dots, K_n$ local reporters. Write $K_{[n]} = (K_1, \dots, K_n)$ and define

$$\partial_J(K_1, \dots, K_n) = \{(s, t) : K_i(s) = K_i(t) \forall i, J(s) \neq J(t)\}. \quad (1)$$

Theorem 1.1 (Factorization biconditional).

$$\partial_J(K_1, \dots, K_n) = \emptyset \iff \exists f \ J = f \circ K_{[n]}. \quad (2)$$

Proof. A factorization makes J constant on every $K_{[n]}$ -fibre. Conversely, if the boundary is empty, define f on each image fibre by the common value of J . $\square$

The exact obstruction is therefore the conjunction of local indistinguishability and global distinguishability.

2 Topological-order specialization

For a degenerate ground-state subspace, local indistinguishability is expressed by

$$\langle \psi_i | O_{\text{local}} | \psi_j \rangle = c(O) \delta_{ij}. \quad (3)$$

Nonlocal loop or logical operators can separate the states. Taking the local expectation-value reporters and a global topological label realizes the reporter-boundary theorem directly.

This exact use must be separated from a looser analogy between diagonal self-reference obstruction and a nontrivial Bloch bundle. A Chern obstruction is an integer cohomology class stable under gap-preserving deformation. The current reflexive obstruction is only the presence of a fixed-point-free twist; it has no proved quantization or deformation stability. Topological theorems cannot be imported solely because both problems involve failure of a global section.

3 Trivial regime

When the relevant obstruction vanishes, a global representation exists. In band theory, under the standard hypotheses, a smooth frame and exponentially localized Wannier basis exist. In the reflexive calculus, the fixed-point regime removes the diagonal contradiction. The shared statement is existence of a global section after the obstruction vanishes; physical consequences remain domain specific.

4 Minimal predictive memory

For a classical stochastic process, causal states are equivalence classes of histories with the same future distribution. Their number is the minimal number of deterministic predictive states. Hidden-state and matrix-product representations require sufficient bond dimension to carry these future-distinguishable conditions.

Quantum models can encode the same classical process in nonorthogonal memory states and can require smaller Hilbert-space dimension than the number of classical causal states. The exact transfer is therefore:

- classical future equivalence gives a deterministic state-count lower bound;
- quantum compression can be smaller when nonorthogonal memory states reproduce the future statistics without being perfectly distinguishable.

5 Joint-state information omitted by marginals

For a bipartite $d_G \times d_Q$ Hermitian joint state, the two normalized marginals omit

$$(d_G^2 - 1)(d_Q^2 - 1) \quad (4)$$

real cross-correlation coordinates. In a 3×3 comparison, 64 joint coordinates are absent. Equal or compatible marginals therefore do not reconstruct the operative relation between them.

6 Obstruction dictionary and grading

The cross-field dictionary is divided into exact, structural, and heuristic entries.

predictive/reflexive object	quantum-materials object	status
reporter boundary	local indistinguishability plus global distinction	exact
minimal deterministic predictive states	classical memory/bond lower bound	exact
fixed-point regime	trivial global-section regime	exact at factorization level
diagonal residual	failure of global section	structural
returned-boundary tower	anomaly inflow hierarchy	structural
typed recurrence depth	renormalization scale	heuristic
record cost	driven-system dissipation	structural; conditional

7 Open reflexive invariant

A topological obstruction is useful because it is graded and stable. The reflexive obstruction currently lacks both properties.

Conjecture 7.1 (Reflexive invariant programme). *There exists a deformation-stable quantity $X(m)$ on a suitably topologized space of self-models that vanishes in the fixed-point regime and grades the residual obstruction otherwise.*

Candidate constructions include a stabilized residual-tower index, an enriched categorical index, or a cohomological lifting obstruction. The conjecture may fail; a no-go theorem would be equally informative.

8 Physical tests

A genuine joint-obstruction claim requires an observable unavailable to every declared local reporter but available to a specified global operation, together with robustness under a declared deformation class. Recommended controls are local tomography, nonlocal loop operators, boundary-mode counting, disorder or perturbation tests, ablation of the global relation, and restoration. Post hoc factorization of two marginals is not such a test.

9 Scope

The factorization biconditional, joint-coordinate count, and classical predictive-state lower bound are exact. The topological dictionary is not uniformly exact. Quantization, stability, bulk-boundary exactness, and the proposed reflexive invariant remain open.

10 Graded translation dictionary

The comparison with quantum materials is intentionally graded. The entries are summarized below.

reporter/representation concept	quantum-materials concept	status
self-model over a state carrier	choice of global smooth frame	structural
fixed-point-free twist	nontrivial holonomy	structural
diagonal residual	obstruction to global section	structural
fixed-point/eigenform regime	topologically trivial phase	exact at the stated existence level
nonfactorization of local reports	local indistinguishability plus global dis- tinction	exact
minimal classical future classes	deterministic bond-dimension lower bound	exact
returned-boundary tower	anomaly-inflow hierarchy	structural; conjectural
typed recurrence depth	renormalization scale	heuristic
dissipation ledger	driven-system heating	structural
formation-weight algebra	effective operator algebra	heuristic

A structural entry transfers a question, not a theorem.

11 Exact reporter-boundary theorem

For reporters $K_i : \Sigma \rightarrow Z_i$ and target label $J : \Sigma \rightarrow O$, define

$$\partial_J(K_1, \dots, K_n) = \{(s, t) : K_i(s) = K_i(t) \ \forall i, J(s) \neq J(t)\}. \quad (5)$$

Then

$$\partial_J = \emptyset \iff J = f \circ (K_1, \dots, K_n) \quad (6)$$

for some f . The forward direction defines f on every joint-reporter fibre, on which J is constant; the reverse direction is immediate. A nonempty boundary is therefore exactly local indistinguishability together with global distinguishability.

In a topologically ordered ground-space, local operators satisfy

$$\langle \psi_i | O_{\text{local}} | \psi_j \rangle = c(O) \delta_{ij}, \quad (7)$$

while nonlocal loop operators separate the states. This is a direct realization of the biconditional.

12 Trivial regime and obstruction limits

When the relevant obstruction vanishes, a global section exists. In band theory this corresponds, under the usual hypotheses, to a smooth frame and localized Wannier representation. In the self-model calculus the

fixed-point regime removes the diagonal contradiction. The analogy stops there unless a deformation-stable invariant is constructed.

The topological obstruction is quantized and stable under gap-preserving deformation. The current reflexive obstruction is only Boolean under the supplied fixed-point-free twist. It has no proved integer index and no specified topology on the space of self-models. Consequently Chern-number, bulk-boundary, and anomaly-inflow theorems cannot be imported as proofs.

13 Memory and bond dimension

For a classical stochastic process, histories with the same future distribution form causal states. Their number is the minimal number of deterministic predictive states. A hidden-state or tensor-network realization must carry enough dimension to distinguish those future conditions. Quantum models can sometimes encode the same classical process in nonorthogonal memory states and use a smaller Hilbert-space dimension. The exact transfer is therefore a classical lower bound plus a possible quantum compression, not equality in every model.

14 Joint information missed by marginals

A normalized Hermitian joint state on $d_G \times d_Q$ has $(d_G d_Q)^2 - 1$ real coordinates. Its two normalized marginals provide only $(d_G^2 - 1) + (d_Q^2 - 1)$. The omitted cross-relation space has dimension

$$(d_G^2 - 1)(d_Q^2 - 1). \quad (8)$$

For a 3×3 comparison, 64 real joint coordinates are absent. Compatible marginals therefore do not reconstruct the operative relation.

15 Open problems and tests

A serious topological extension must define a topology or deformation class on self-models, construct an invariant that vanishes in the fixed-point regime, prove stability, and derive a boundary observable. Physical tests should compare local tomography with a specified nonlocal operation, perform perturbation and disorder tests, and use ablation/restoration of the global relation. Post hoc factorization of two marginals does not meet this standard.

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